



WiMAX equipment

enhanced support for femtocells, introduction of relay stations, faster handovers, lower media access control (MAC) overhead, more efficient power-saving modes, enhanced multicast-broadcast services (MBS) and location-based services (LBS). Other enhancements include a slightly higher spectral efficiency, lower link access latencies, emphasis on both TDD and frequency division duplexing (FDD) operations, multi-radio coexistence and inter-technology handover, integrated multi-hop relay, better security, self-organising and self-optimising base stations, additional target frequencies of operations, and so on.

It is claimed that 802.16m is completely backwards-compatible with its predecessor. Hence, although 802.16m requires new hardware that meets the specifications, 802.16e equipment can inter-operate with 802.16m equipment. This ensures smooth migration paths for 802.16e operators and inter-operability in roaming between both types of operators. However, some experts feel that the handover capabilities in 802.16e WiMAX are somewhat unproven and the migration path from 802.16e to 802.16m WiMAX is also somewhat questionable although not disruptive.

3GPP LTE Release 10: Who will win the race?

The existing LTE standard—Release 8—is considered a pre-4G one. That is, it addresses a stage in the progression from 3G UMTS to 4G. However, forerunning implementations of Release 8, such as those by TeliaSonera in Sweden as well as AT&T and Verizon in the US, have been quite successful and so the bars are set quite high for the upcoming full-4G version—LTE Release 10 (also called LTE-Advanced). This is expected to be WiMAX's biggest competitor, and many experts are betting on it winning the race, while others think the two will coexist for the consumer's good.

"LTE meets, and in most cases exceeds, the requirements for a 4G technology. Through a range of innovative functionalities, LTE enables operators to manage more traffic and meet growing data-rate demands, and is consequently a key enabler for future mobile broadband delivery," says Dhawan, adding that LTE-Advanced also supports speeds of 1 Gbps or more.

LTE is an OFDM-based radio-access technology that supports up to 20MHz scalable transmission bandwidth and advanced multi-antenna transmission including beam-forming and spatial

Quick look: WCDMA vs OFDM technologies

"Clearly WCDMA is less advanced and less efficient compared to newer OFDM technologies like WiMAX and LTE," says Shyam Ananthanarayan. Here is why...

1. WCDMA technology is now able to give only up to 16 quadrature amplitude modulation or QAM (4 bits per symbol) whereas LTE based on OFDM already delivers 64 QAM (6 bits per symbol) and soon will deliver 256 QAM (8 bits per symbol). That means OFDM clearly delivers more bits per hertz of spectrum.
2. Most modulation schemes lose bits in error correction schemes. Here also OFDM beats WCDMA.
3. OFDM-based technologies incorporate other advanced techniques like MIMO, smart antenna, sub-channelisation, etc, which increase the efficiency of these modulations in delivering high throughputs compared to WCDMA.
4. WCDMA involves chip-rate processing where every symbol will have to be multiplied by a large number, as large as 3,840,000 chips, requiring huge processing hardware, making the equipment and handsets power hungry. That means your battery drains faster, makes them bulkier and more expensive.
5. The base stations for current WCDMA sites use up to 6000 watt of power, whereas one can build LTE base stations that use 500 watt or less. In many emerging countries, this 10X advantage in power consumption makes a huge difference because most cell-sites take up huge cost in air-conditioner housing, generators, battery backup, power management systems, etc.

multiplexing, with up to four transmit antennas in the downlink direction. "LTE uses OFDM access for downlink but single-carrier FDMA (SC-FDMA) for uplink," adds Dr Borkar.

Like 802.16m WiMAX, LTE also has a single cell geographical coverage subject to the same frequency of operation, simplified flat packet transport based architectures, adaptive modulation to maintain consistent bit error rate, hybrid-automatic request and cyclic prefixing for handling transmission errors, and so on. However, while